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SOIL CORROSIVITY TESTING APPARATUS

Abstract

Implementations of the present disclosure includes a testing apparatus that includes a base, multiple threaded studs, and a plate. The base has a body defining a flat surface and multiple apertures each arranged to receive an electrode extending through a respective one of the apertures into soil disposed beneath the base. The threaded studs are rotationally coupled to the base. The plate is threadedly coupled to the threaded studs such that rotation of the threaded studs changes an elevation of the plate with respect to the base. The plate includes a lowermost surface facing the base and an uppermost surface facing away from the base and arranged to receive a load. The lowermost surface is coupled to an end of each electrode such that, as the plate changes in elevation, each electrode moves through its respective aperture, changing a position of each electrode with respect to the soil.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates soil corrosivity testing systems.

BACKGROUND

[0002] Soil can be tested for corrosivity to determine the extent of corrosive soil and corrosive water for oil production purposes. Three-electrode electrochemical systems use electrochemical reactions to measure the concentration of chemicals in the soil and thereby determine the corrosivity of the soil. Methods and equipment for improving corrosivity testing are sought.

SUMMARY

[0003] Implementations of the present disclosure include a testing assembly that includes a beaker, a sample holder, soil, a sample specimen, a three-electrode testing assembly, an electrochemical analyzer, and a laser assembly. The sample holder is disposed at a bottom surface of the beaker. The soil is disposed within the beaker and on the sample holder. The sample specimen is disposed on the sample holder. The three-electrode testing assembly is disposed partially within the beaker and on the soil. The three-electrode testing assembly includes a base, multiple threaded knobs, and a plate. The base has a body defining a flat surface arranged to bear against and compact the soil. The base defines multiple apertures each arranged to receive an electrode extending through a respective one of the apertures into the soil. The threaded knobs are rotationally coupled to the base and extend from the base normal with respect to the flat surface of the base. The plate is threadedly coupled to the threaded knobs such that rotation of the threaded knobs changes an elevation of the plate with respect to the base. The plate has a lowermost surface facing the base and an uppermost surface facing away from the base and arranged to receive a load. The lowermost surface is coupled to an end of each electrode such that, as the plate changes in elevation with respect to the base, each electrode moves through its respective aperture, changing a position of each electrode with respect to the soil. The electrochemical analyzer is electrically coupled to each electrode and processes information received from each electrode and determines, as a function of the information, an electrochemical corrosion of the sample specimen. The laser assembly includes a laser emitter and a laser reflector. The laser emitter is coupled to an outer wall of the beaker and includes an electronic display. The laser reflector is coupled to the plate and is aligned with the laser emitter such that the laser reflector reflects a laser emitted from the laser emitter to the laser emitter, allowing the laser emitter toa allow the emitter to detect the reflected laser and determine, as a function of the reflected laser, a distance between the laser emitter and the laser reflector.

[0004] In some implementations, the laser emitter includes a microcontroller, an electronic display, and a power source configured to power the electronic display. The microcontroller determines the distance as the distance changes as the three-electrode testing assembly compacts the soil under the load. The electronic display displays the distance in real time. In some implementations, the microcontroller determines, as a function of the determined distance, when a reduction in distance stabilizes, indicating that compaction is completed. The laser reflector further includes a speaker that emits a sound when the microcontroller determines that compaction is completed.

[0005] In some implementations, the three-electrode testing assembly includes a ground-penetrating radar (GPR) coupled to the base. The GPR generates and transmit subsurface images to a processing device and the processing device determines, as a function of the subsurface images, at least one of a compaction level of the soil or a moisture level of the soil.

[0006] In some implementations, the plate includes an indentation at a center of the plate to support a weight applying the load and prevent the weight from shifting along the uppermost surface of the plate, allowing the load to be applied uniformly over a period of compaction.

[0007] Implementations of the present disclosure includes a testing apparatus that includes a base, multiple threaded studs, and a plate. The base has a body defining a flat surface and multiple apertures each arranged to receive an electrode extending through a respective one of the apertures into soil disposed beneath the base. The threaded studs are rotationally coupled to the base and extend from the base normal with respect to the flat surface of the base. The plate is threadedly coupled to the threaded studs such that, with the testing apparatus assembled, rotation of the threaded studs changes an elevation of the plate with respect to the base. The plate includes a lowermost surface facing the base and an uppermost surface facing away from the base and arranged to receive a load. The lowermost surface is arranged to be coupled to an end of each electrode such that, as the plate changes in elevation with respect to the base, each electrode moves through its respective aperture, changing a position of each electrode with respect to the soil.

[0008] In some implementations, the testing apparatus further includes a beaker configured to hold a sample with the soil inside the beaker and on the sample. The body of the base defines a flat surface arranged to bear against and compact the soil.

[0009] In some implementations, the body includes a circular body defining a diameter corresponding with an inner diameter of the beaker such that the circular body is movable, with the flat surface parallel with respect to a base of the beaker, along the beaker to allow the circular body to move toward the base of the beaker and compact, under the load, the soil between the sample, the flat surface, and an inner wall of the beaker.

[0010] In some implementations, the testing apparatus further includes a sample holder arranged to rest on the base of the beaker and includes an indentation that holds the sample and prevents the sample from shifting along an upper surface of the sample holder.

[0011] In some implementations, the testing apparatus includes a three-electrode testing assembly. Each electrode is electrically coupled to an electrochemical analyzer configured to process information received from each electrode and determine, as a function of the information, an electrochemical corrosion of the sample.

[0012] In some implementations, the apertures include four or more apertures, including a first aperture configured to receive a reference electrode, a second aperture configured to receive a working electrode, a third aperture configure to receive a counter electrode, and a further aperture configured to receive an instrument including at least one of a thermocouple or a pH measurement instrument.

[0013] In some implementations, the testing apparatus also includes a laser assembly that includes a laser emitter arranged to be coupled to an outer wall of the beaker and include an electronic display. The laser reflector is coupled to the plate and aligned with the laser emitter such that a laser emitted from the laser emitter reflects back to the laser emitter, allowing the laser emitter to detect the reflected laser and determine, as a function of the reflected laser, a distance between the laser emitter and the laser reflector to determine a distance that the soil compacts under the load.

[0014] In some implementations, the laser emitter includes a microcontroller, an electronic display, and a power source configured to power the electronic display. The microcontroller determines the distance as the distance changes as the three electrode testing assembly compacts the soil under the load. The electronic display displays the distance in real time.

[0015] In some implementations, the microcontroller determines, as a function of the determined distance, when a reduction in distance stabilizes, indicating that compaction is completed. The laser reflector further includes a speaker configured it emit a sound when the compaction is completed.

[0016] In some implementations, the testing assembly further includes a ground-penetrating radar (GPR) coupled to the base, the GPR generates and transmits subsurface images to a processing device and the processing device determines, as a function of the subsurface images, at least one of a compaction level of the soil or a moisture level of the soil.

[0017] In some implementations, the plate includes an indentation at a center of the plate to support a weight applying the load and prevent the weight from shifting along the uppermost surface of the plate, allowing the load to be applied uniformly over a period of compaction.

[0018] In some implementations, rotation of the studs changes an elevation of the plate without changing an elevation of the screws with respect to the base.
[0019] In some implementations, the threaded studs include threaded knobs, each of the threaded knobs spaced apart equidistantly from one another.
[0020] Implementations of the present disclosure include a method that includes obtaining a testing apparatus that includes a base, threaded studs, and a plate. The base includes a body defining a flat surface and apertures each arranged to receive an electrode extending through a respective one of the apertures into soil disposed beneath the base. The threaded studs are rotationally coupled to the base and extend from the base normal with respect to the flat surface of the base. The plate is threadedly coupled to the studs such that rotation of the studs changes an elevation of the plate with respect to the base. The plate includes a lowermost surface facing the base and coupled to an end of each electrode such that, as the plate changes in elevation with respect to the base, each electrode moves through its respective aperture, changing a position of each electrode with respect to the soil. The method also includes placing a sample specimen inside a beaker. The method also includes placing the soil inside the beaker and on the sample specimen. The method also includes placing the testing apparatus on the soil with at least one electrode extending into the soil. The method also includes placing a weight on the plate to apply a load on the testing apparatus, allowing the testing apparatus to compact the soil under the load as each electrode detects a parameter of the sample specimen or the soil.
[0021] In some implementations, the method further includes including rotating the threaded studs as the testing apparatus compacts the soil, moving each electrode with respect to the base to maintain a desired distance between a tip of each electrode and the sample specimen.
[0022] Particular implementations of the subject matter described in this specification can be implemented so as to realize one or more of the following advantages. For example, the testing assembly of the present disclosure allows compaction of the soil without requiring manual compaction, and while the testing apparatus tests for soil corrosivity. Also, the testing apparatus allows consistent compaction of the soil, which can result in more accurate testing results. Moreover, the testing apparatus allows the accurate monitoring and control of the compaction process, ensuring uniformity and consistency across multiple experimental trials. In addition, the laser feature offers a wider range of compact levels, allowing for a more accurate simulation of actual soil conditions found in the field. This can increase the accuracy and consistency of laboratory tests, broaden the scope and utility of the testing apparatus, and allow a wider range of soil conditions to be accurately tested.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0023] FIG. **1** shows front schematic view of a testing assembly.
[0024] FIG. **2** shows a front perspective view of a testing apparatus of the testing assembly.
[0025] FIG. **3** shows a top view of the testing apparatus.
[0026] FIG. **4** shows a top schematic view of a sample holder of the testing assembly.
[0027] FIG. **5** show a flow chart of an example method of testing soil for corrosivity.
DETAILED DESCRIPTION OF THE DISCLOSURE
[0028] FIG. **1** shows a compaction and testing assembly **100** (e.g., a three-electrode system or soil corrosivity evaluation device) that includes a beaker **102**, a sample holder **104**, a sample **108** (e.g., a metal specimen or coupon), soil **106**, a testing apparatus **110** (e.g., a three-electrode cell or testing assembly), and a laser assembly **130**. The testing assembly **100** can perform a corrosivity test while at the same time compact the soil uniformly.
[0029] In some aspects, the sample holder **104** is disposed at a bottom surface **103** (e.g., the base)

of the beaker **102**. The sample **108** rests on the sample holder **104**. During the corrosivity test, the soil **106** resides within the beaker **102** between the sample holder **104** and the testing apparatus **110**. During the test, the testing apparatus **110** resides partially within the beaker **102** and sits on the soil **106** to compress the soil **106**.

[0030] The testing apparatus **110** includes a base **112**, multiple threaded studs **120** (e.g., threaded knobs with handles **121**), and a plate **122**. The base **112** of the testing apparatus **110** has a circular body **114** that defines a flat surface **116** that bears against and compacts the soil **106**. The body has multiple apertures **119** each arranged to receive an electrode **118** (shown in dashed lines for clarity) extending through a respective one of the apertures **119** into the soil **106**. For example, one of the apertures **119** (e.g., the central aperture) receives a reference electrode **118***a*, a second aperture **119** receives a working electrode **118***b*, and a third aperture **119** receives a counter electrode **118***c*. As further described in detail below with respect to FIG. **2**. the circular body **114** can have more than three holes **119** to accommodate more instruments.

[0031] In some aspects, the testing apparatus **110** also includes an electrochemical analyzer **128** electrically coupled to each electrode **118**. The electrochemical analyzer **128** processes information received from each electrode **118** and determines, as a function of the information, the corrosivity of the soil **106** or the electrochemical corrosion of the sample **108**. In some aspects, the electrochemical analyzer **128** also determines other parameters of the sample or soil or both, such as the concentration of various chemicals in the soil.

[0032] The threaded studs **120** are rotationally coupled to the base **112** and extend from the base **112** in a direction normal (e.g., perpendicular) with respect to the flat surface **116** of the base **112**. The threaded studs are attached (e.g., with a rotational bearing attachment **127**) to the body **114** of the base **112** to allow rotation of the studs **120** without screwing in or out the screws from the body **114**.

[0033] The plate **122** as threaded holes **131** that correspond to the locations of each threaded stud **120**. The threaded holes **131** extend through the plate **122** and allow the plate **122** to be threadedly attached to each threaded stud **120** such that rotation of the threaded studs **120** changes an elevation of the plate **122** with respect to the base **112**. The plate **122** has a lowermost surface **124** facing the base **112** and an uppermost surface **126** facing away from the base **112**. The uppermost surface **126** supports a weight **128** that applies a load on the testing apparatus **110** to compact the soil **106**. The lowermost surface **124** is coupled (e.g., through a mechanical fastener **113** or an adhesive or the like) to an end of each electrode **118** such that, as the plate **122** changes in elevation with respect to the base **111**, each electrode **118** moves along its respective aperture **119**, changing a position (e.g., the depth) of each electrode **118** within the soil **116**.

[0034] In some aspects, the threaded studs **120** are rotatable to lift the plate **122** with respect to the base **112** as the soil is compacted, allowing an operator to prevent the electrodes **118** from touching the bottom surface **103** of the beaker **102**. For example, to keep the reference electrode **118***a* facing the sample **108** as close as possible (and thus ensure appropriate electrochemical measurements), an operator can rotate, as the soil **106** is compacted, the threaded studs **120** to lift the plate **122** and thereby the electrodes **118**, maintaining the reference electrode **118** at a desired distance from the sample **108**.

[0035] The laser assembly **130** has a laser emitter **132** (e.g., a transceiver) and a laser reflector **134**. In some aspects, the laser emitter **132** is attached to an outer wall of the beaker **102**. For example, the laser reflector **134** can be attached to the beaker **102** with an adhesive, a mechanical fastener, or the like. The laser reflector **134** is coupled to the plate **122** and is vertically aligned with the laser emitter **132** such that a laser **111** emitted from the laser emitter **132** reflects back from the reflector **132** to the laser emitter **132**. This allows the laser emitter **132** to detect the reflected laser **111** and determine, as a function of the reflected laser **111**, a distance between the laser emitter **132** and the laser reflector **134**.

[0036] In some aspects, the laser emitter **132** has an electronic display **132** (e.g., a digital screen), a power source **138** (e.g., a battery), a microcontroller **135**, and a speaker **140**. The power source **138** powers the electronic display **136**. The microcontroller **135** determines, as a function of the emitter detecting the reflected laser, the distance between the laser emitter **132** and the laser reflector **134**. For example, the microcontroller **135** has a processor that receives information form a receiver of the emitter **132** and that the receiver transmits to the processor in response to receiving or sensing the reflected laser. In some aspects, the microcontroller **138** determines, in real time, the distance as the distance changes during compaction of the soil **106**. The electronic display **136** displays the distance as the distance chances. For example, the electronic display **126** display the distance in real time as the distance between the plate **122** and the emitter **132** decreases because of the soil's compaction under the weight **128**.

[0037] In some aspects, the microcontroller **138** determines, as a function of the determined distance, when a reduction in distance has stabilized, which indicates that compaction is completed. For example, the microcontroller **138** can compare, to a time period threshold, the time period that it takes for the distance to change. Them, the microcontroller **138** determines that the compaction is complete when the determined time period satisfies the threshold. For example, if the threshold is **60** seconds, the microcontroller **138** determines, once the distance remains unchanged for **60** seconds, that the soil **106** has been fully compacted under the load **128**. In some aspects, the laser emitter **132** has a speaker connected to and controllable by the the microcontroller **138** to emit a sound when the compaction is completed.

[0038] In some aspects, the laser assembly **130** determines the soil compaction with high precision. This feature allows for accurate monitoring and control of the compaction process, ensuring uniformity and consistency across multiple experimental trials. By precisely measuring the reduction in distance, researchers can maintain standardized soil conditions, minimizing variations and external influences that could affect the electrochemical testing.

[0039] Moreover, the laser assembly **130** can be used as a leveling mechanism to ensure horizontal leveling of compaction. For example, as shown in FIG. **1**, the apparatus can incorporate two pairs of laser emitter and reflector assemblies, one on each side of the beaker **102**. If both laser emitters **132** detect the same distance, the laser assembly **130** indicates that the compaction is horizontally leveled. However, if the distances differ, it signifies an uneven compaction that requires adjustment. This allows the electrolyte (e.g., the soil) to distribute evenly. If the soil **106** has a tilted surface, the test can be repeater or adjusted.

[0040] In some aspects, the testing apparatus **110** has a ground-penetrating radar (GPR) **139** attached to the base **112**. The GPR **139** generates subsurface images of the soil **106** and transmits the images to a processing device **141** which can be past of (or the same as) the electrochemical analyzer **128** or a different processing device. The processing device **141** determines, as a function of the subsurface images, a compaction level of the soil **106** and/or a moisture level of the soil **106**.

[0041] In some aspects, the GPR **139** has a transmitting and receiving antennas that emit electromagnetic waves at specific frequencies and receive the reflected signals after the waves penetrated the soil **106** and interacted with subsurface objects or boundaries. By acquiring radar data from the field, it becomes possible to replicate soil compaction and moisture levels conditions in a laboratory. The radar data serves as a reference for achieving the desired compaction and moisture levels when conducting experiments or tests in a controlled lab environment. This allows realistic field conditions to be replicated, enhancing the reliability and applicability of laboratory research. Thus, the integration of GRP **139** with the laser assembly **130** provides a comprehensive approach to replicate field soil conditions accurately.

[0042] FIG. **2** shows a front perspective view of the testing apparatus **110**. Referring also to FIG. **1**, the body **114** of the base **112** is circular and has a diameter that corresponds with (e.g., is slightly smaller than) an inner diameter of the beaker **102**. This allows that the circular body **114** move, with its flat surface **116** parallel with respect to the bottom surface **103** of the beaker **102**, along the height of beaker **102**. This allows the circular body **114** to move toward the bottom surface **103** of the beaker **102** to compact, under the load of the weight **128**, the soil **106** between the sample holder **104**, the flat surface **116** of the base **114**, and the inner wall of the beaker **102**.

[0043] In some aspects, the circular body **114** has more than three apertures **119** (e.g., seven apertures) of the same or different sizes. Each aperture receives an instrument that detects a parameter of the soil or the sample. For example, the additional apertures **119** can receive a thermocouple or a pH measurement instrument. Also, as shown in FIG. **2**, the threaded studs **120** are spaced apart equidistantly from one another so that the load is evenly distributed among the threaded studs **120**.

[0044] FIG. **3** shows a top view of the testing apparatus **110**. As shown, the plate **122** has a feature that keeps the weight from shifting along the top surface of the plate **122**. For example, the plate **122** has an indentation **123** (e.g., a circular groove) at its center to receive the weight **128** and prevent the weight **128** from substantially shifting along the uppermost surface of the plate **122**. This allows the load applied by the weight **128** to be applied uniformly on the soil **106** over the period of compaction.

[0045] FIG. **4** shows a perspective view of the sample holder **104**. Similar to the circular body **122**, the sample holder **104** has a diameter that corresponds with (e.g., is slightly smaller than) an inner diameter of the beaker **102**. This allows the majority of the sand **106** to rest on the topo surface of the holder **104**, preventing substantial soil from falling through the side of the holder **104**. In some aspects, the sample holder **104** has an indentation **109** (e.g., a groove) that holds the sample **108** and prevents the sample **108** from shifting along an upper surface of the sample holder **104**.

[0046] FIG. **5** shows a flow chart of a method (**500**) of testing soil for corrosivity. The method includes obtaining a testing assembly (**505**). The testing assembly can be the testing assembly **100** described above with respect to FIGS. **1-4**. The method also includes placing a sample specimen inside a beaker (**510**). The method also includes placing the soil inside the beaker and on the sample specimen (**515**). The method also includes placing the testing apparatus on the soil with at least one electrode extending into the soil (**520**). The method also includes placing a weight on the plate to apply a load on the testing apparatus, allowing the testing apparatus to compact the soil under the load as each electrode detects a parameter of the sample specimen or the soil (**525**).

[0047] While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any inventions or of what may be claimed, but rather as descriptions of features specific to particular implementations of particular inventions. Certain features that are described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

[0048] Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

[0049] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. For example, example operations, methods, or processes described herein may include more steps or fewer steps than those described. Further, the steps in such example operations, methods, or processes may be performed in different successions than that described or illustrated in the figures. Accordingly, other implementations are within the scope of the following claims.

EXAMPLES

[0050] In an example implementation, a testing assembly includes a beaker, a sample holder, soil, a sample specimen, a three-electrode testing assembly, an electrochemical analyzer, and a laser assembly. The sample holder is disposed at a bottom surface of the beaker. The soil is disposed within the beaker and on the sample holder. The sample specimen is disposed on the sample holder. The three-electrode testing assembly is disposed partially within the beaker and on the soil. The three-electrode testing assembly includes a base, multiple threaded knobs, and a plate. The base has a body defining a flat surface arranged to bear against and compact the soil. The base defines multiple apertures each arranged to receive an electrode extending through a respective one of the apertures into the soil. The threaded knobs are rotationally coupled to the base and extend from the base normal with respect to the flat surface of the base. The plate is threadedly coupled to the threaded knobs such that rotation of the threaded knobs changes an elevation of the plate with respect to the base. The plate has a lowermost surface facing the base and an uppermost surface facing away from the base and arranged to receive a load. The lowermost surface is coupled to an end of each electrode such that, as the plate changes in elevation with respect to the base, each electrode moves through its respective aperture, changing a position of each electrode with respect to the soil. The electrochemical analyzer is electrically coupled to each electrode and processes information received from each electrode and determines, as a function of the information, an electrochemical corrosion of the sample specimen. The laser assembly includes a laser emitter and a laser reflector. The laser emitter is coupled to an outer wall of the beaker and includes an electronic display. The laser reflector is coupled to the plate and is aligned with the laser emitter such that the laser reflector reflects a laser emitted from the laser emitter to the laser emitter, allowing the laser emitter toa allow the emitter to detect the reflected laser and determine, as a function of the reflected laser, a distance between the laser emitter and the laser reflector.

[0051] In an example implementation combinable with any other example implementation, the laser emitter includes a microcontroller, an electronic display, and a power source configured to power the electronic display. The microcontroller determines the distance as the distance changes as the three-electrode testing assembly compacts the soil under the load. The electronic display displays the distance in real time.

[0052] In an example implementation combinable with any other example implementation, the microcontroller determines, as a function of the determined distance, when a reduction in distance stabilizes, indicating that compaction is completed. The laser reflector further includes a speaker that emits a sound when the microcontroller determines that compaction is completed.

[0053] In an example implementation combinable with any other example implementation, the three-electrode testing assembly includes a ground-penetrating radar (GPR) coupled to the base. The GPR generates and transmit subsurface images to a processing device and the processing device determines, as a function of the subsurface images, at least one of a compaction level of the soil or a moisture level of the soil.

[0054] In an example implementation combinable with any other example implementation, the plate includes an indentation at a center of the plate to support a weight applying the load and prevent the weight from shifting along the uppermost surface of the plate, allowing the load to be applied uniformly over a period of compaction.

[0055] In an example implementation, a testing apparatus includes a base, multiple threaded studs, and a plate. The base has a body defining a flat surface and multiple apertures each arranged to receive an electrode extending through a respective one of the apertures into soil disposed beneath the base. The threaded studs are rotationally coupled to the base and extend from the base normal with respect to the flat surface of the base. The plate is threadedly coupled to the threaded studs such that, with the testing apparatus assembled, rotation of the threaded studs changes an elevation

of the plate with respect to the base. The plate includes a lowermost surface facing the base and an uppermost surface facing away from the base and arranged to receive a load. The lowermost surface is arranged to be coupled to an end of each electrode such that, as the plate changes in elevation with respect to the base, each electrode moves through its respective aperture, changing a position of each electrode with respect to the soil.
[0056] In an example implementation combinable with any other example implementation, the testing apparatus further includes a beaker configured to hold a sample with the soil inside the beaker and on the sample. The body of the base defines a flat surface arranged to bear against and compact the soil.
[0057] In an example implementation combinable with any other example implementation, the body includes a circular body defining a diameter corresponding with an inner diameter of the beaker such that the circular body is movable, with the flat surface parallel with respect to a base of the beaker, along the beaker to allow the circular body to move toward the base of the beaker and compact, under the load, the soil between the sample, the flat surface, and an inner wall of the beaker.
[0058] In an example implementation combinable with any other example implementation, the testing apparatus further includes a sample holder arranged to rest on the base of the beaker and includes an indentation that holds the sample and prevents the sample from shifting along an upper surface of the sample holder.
[0059] In an example implementation combinable with any other example implementation, the testing apparatus includes a three-electrode testing assembly. Each electrode is electrically coupled to an electrochemical analyzer configured to process information received from each electrode and determine, as a function of the information, an electrochemical corrosion of the sample.
[0060] In an example implementation combinable with any other example implementation, the apertures include four or more apertures, including a first aperture configured to receive a reference electrode, a second aperture configured to receive a working electrode, a third aperture configure to receive a counter electrode, and a further aperture configured to receive an instrument including at least one of a thermocouple or a pH measurement instrument.
[0061] In an example implementation combinable with any other example implementation, the testing apparatus also includes a laser assembly that includes a laser emitter arranged to be coupled to an outer wall of the beaker and include an electronic display. The laser reflector is coupled to the plate and aligned with the laser emitter such that a laser emitted from the laser emitter reflects back to the laser emitter, allowing the laser emitter to detect the reflected laser and determine, as a function of the reflected laser, a distance between the laser emitter and the laser reflector to determine a distance that the soil compacts under the load.
[0062] In an example implementation combinable with any other example implementation, the laser emitter includes a microcontroller, an electronic display, and a power source configured to power the electronic display. The microcontroller determines the distance as the distance changes as the three electrode testing assembly compacts the soil under the load. The electronic display displays the distance in real time.
[0063] In an example implementation combinable with any other example implementation, the microcontroller determines, as a function of the determined distance, when a reduction in distance stabilizes, indicating that compaction is completed. The laser reflector further includes a speaker configured it emit a sound when the compaction is completed.
[0064] In an example implementation combinable with any other example implementation, the testing assembly further includes a ground-penetrating radar (GPR) coupled to the base, the GPR generates and transmits subsurface images to a processing device and the processing device determines, as a function of the subsurface images, at least one of a compaction level of the soil or a moisture level of the soil.
[0065] In an example implementation combinable with any other example implementation, the

plate includes an indentation at a center of the plate to support a weight applying the load and prevent the weight from shifting along the uppermost surface of the plate, allowing the load to be applied uniformly over a period of compaction.

[0066] In an example implementation combinable with any other example implementation, rotation of the studs changes an elevation of the plate without changing an elevation of the screws with respect to the base.

[0067] In an example implementation combinable with any other example implementation, the threaded studs include threaded knobs, each of the threaded knobs spaced apart equidistantly from one another.

[0068] In an example implementation, a method includes obtaining a testing apparatus that includes a base, threaded studs, and a plate. The base includes a body defining a flat surface and apertures each arranged to receive an electrode extending through a respective one of the apertures into soil disposed beneath the base. The threaded studs are rotationally coupled to the base and extend from the base normal with respect to the flat surface of the base. The plate is threadedly coupled to the studs such that rotation of the studs changes an elevation of the plate with respect to the base. The plate includes a lowermost surface facing the base and coupled to an end of each electrode such that, as the plate changes in elevation with respect to the base, each electrode moves through its respective aperture, changing a position of each electrode with respect to the soil. The method also includes placing a sample specimen inside a beaker. The method also includes placing the soil inside the beaker and on the sample specimen. The method also includes placing the testing apparatus on the soil with at least one electrode extending into the soil. The method also includes placing a weight on the plate to apply a load on the testing apparatus, allowing the testing apparatus to compact the soil under the load as each electrode detects a parameter of the sample specimen or the soil.

[0069] In an example implementation combinable with any other example implementation, the method further includes including rotating the threaded studs as the testing apparatus compacts the soil, moving each electrode with respect to the base to maintain a desired distance between a tip of each electrode and the sample specimen.

Claims

1. A testing assembly, comprising: a beaker; a sample holder disposed at a bottom surface of the beaker; soil disposed within the beaker and on the sample holder; a sample specimen disposed on the sample holder; a three-electrode testing assembly disposed partially within the beaker and on the soil, the three-electrode testing assembly comprising: a base comprising a body defining a flat surface arranged to bear against and compact the soil, the base defining a plurality of apertures each arranged to receive an electrode extending through a respective one of the plurality of apertures into the soil; a plurality of threaded knobs rotationally coupled to the base and extend from the base normal with respect to the flat surface of the base; and a plate configured to be threadedly coupled to the plurality of threaded knobs such that rotation of the plurality of threaded knobs changes an elevation of the plate with respect to the base, the plate comprising a lowermost surface facing the base and an uppermost surface facing away from the base and arranged to receive a load, wherein the lowermost surface is coupled to an end of each electrode such that, as the plate changes in elevation with respect to the base, each electrode moves through its respective aperture, changing a position of each electrode with respect to the soil; an electrochemical analyzer electrically coupled to each electrode and configured to process information received from each electrode and determine, as a function of the information, an electrochemical corrosion of the sample specimen; and a laser assembly, comprising: a laser emitter coupled to an outer wall of the beaker and comprising an electronic display; and a laser reflector coupled to the plate and aligned with the laser emitter such that a laser emitted from the laser emitter reflects back to the laser emitter, allowing the laser emitter to detect the reflected laser and determine, as a function of the reflected laser, a distance between the laser emitter and the laser reflector.

2. The testing assembly of claim 1, wherein the laser emitter comprises a microcontroller, an electronic display, and a power source configured to power the electronic display, the microcontroller configured to determine the distance as the distance changes as the three-electrode testing assembly compacts the soil under the load, and the electronic display configured to display the distance in real time.

3. The testing assembly of claim 2, wherein the microcontroller is configured to determine, as a function of the determined distance, when a reduction in distance stabilizes, indicating that compaction is completed, and the laser reflector further comprises a speaker configured it emit a sound when the compaction is completed.

4. The testing assembly of claim 1, wherein the three-electrode testing assembly comprises a ground-penetrating radar (GPR) coupled to the base, the GPR configured to generate and transmit subsurface images to a processing device and the processing device is configured to determine, as a function of the subsurface images, at least one of a compaction level of the soil or a moisture level of the soil.

5. The testing assembly of claim 1, wherein the plate comprises an indentation at a center of the plate to support a weight applying the load and prevent the weight from shifting along the uppermost surface of the plate, allowing the load to be applied uniformly over a period of compaction.

6. A testing apparatus, comprising: a base comprising a body defining a flat surface and a plurality of apertures each arranged to receive an electrode extending through a respective one of the plurality of apertures into soil disposed beneath the base; a plurality of threaded studs configured to be rotationally coupled to the base and extend from the base normal with respect to the flat surface of the base; and a plate configured to be threadedly coupled to the plurality of threaded studs such that, with the testing apparatus assembled, rotation of the plurality of threaded studs changes an elevation of the plate with respect to the base, the plate comprising a lowermost surface facing the base and an uppermost surface facing away from the base and arranged to receive a load, wherein the lowermost surface is arranged to be coupled to an end of each electrode such that, as the plate changes in elevation with respect to the base, each electrode moves through its respective aperture, changing a position of each electrode with respect to the soil.

7. The testing apparatus of claim 6, further comprising a beaker configured to hold a sample with the soil inside the beaker and on the sample, the body of the base defining a flat surface arranged to bear against and compact the soil.

8. The testing apparatus of claim 7, wherein the body comprises a circular body defining a diameter corresponding with an inner diameter of the beaker such that the circular body is movable, with the flat surface parallel with respect to a base of the beaker, along the beaker to allow the circular body to move toward the base of the beaker and compact, under the load, the soil between the sample, the flat surface, and an inner wall of the beaker.

9. The testing apparatus of claim 8, further comprising a sample holder arranged to rest on the base of the beaker and comprising an indentation that holds the sample and prevents the sample from shifting along an upper surface of the sample holder.

10. The testing apparatus of claim 7, wherein the testing apparatus comprises a three-electrode testing assembly, and each electrode is electrically coupled to an electrochemical analyzer configured to process information received from each electrode and determine, as a function of the information, an electrochemical corrosion of the sample.

11. The testing apparatus of claim 10, wherein the plurality of apertures comprises four or more apertures, comprising a first aperture configured to receive a reference electrode, a second aperture configured to receive a working electrode, a third aperture configure to receive a counter electrode, and a further aperture configured to receive an instrument comprising at least one of a thermocouple or a pH measurement instrument.

12. The testing apparatus of claim 7, further comprising a laser assembly, comprising: a laser emitter arranged to be coupled to an outer wall of the beaker and comprising an electronic display; and a laser reflector arranged to be coupled to the plate and aligned with the laser emitter such that a laser emitted from the laser emitter reflects back to the laser emitter, allowing the laser emitter to detect the reflected laser and determine, as a function of the reflected laser, a distance between the laser emitter and the laser reflector to determine a distance that the soil compacts under the load.

13. The testing apparatus of claim 12, wherein the laser emitter comprises a microcontroller, an electronic display, and a power source configured to power the electronic display, the microcontroller configured to determine the distance as the distance changes as the three electrode testing assembly compacts the soil under the load, and the electronic display configured to display the distance in real time.

14. The testing apparatus of claim 12, wherein the microcontroller is configured to determine, as a function of the determined distance, when a reduction in distance stabilizes, indicating that compaction is completed, and the laser reflector further comprises a speaker configured it emit a sound when the compaction is completed.

15. The testing apparatus of claim 6, further comprising a ground-penetrating radar (GPR) coupled to the base, the GPR configured to generate and transmit subsurface images to a processing device and the processing device is configured to determine, as a function of the subsurface images, at least one of a compaction level of the soil or a moisture level of the soil.

16. The testing apparatus of claim 6, wherein the plate comprises an indentation at a center of the plate to support a weight applying the load and prevent the weight from shifting along the uppermost surface of the plate, allowing the load to be applied uniformly over a period of compaction.

17. The testing apparatus of claim 6, wherein rotation of the studs changes an elevation of the plate without changing an elevation of the screws with respect to the base.

18. The testing apparatus of claim 6, wherein the plurality of threaded studs comprises a plurality of threaded knobs, each of the plurality of threaded knobs spaced apart equidistantly from one another.

19. A method, comprising: obtaining a testing apparatus comprising: a base comprising a body defining a flat surface and a plurality of apertures each arranged to receive an electrode extending through a respective one of the plurality of apertures into soil disposed beneath the base; a plurality of threaded studs rotationally coupled to the base and extending from the base normal with respect to the flat surface of the base; and a plate threadedly coupled to the plurality of studs such that rotation of the plurality of studs changes an elevation of the plate with respect to the base, the plate comprising a lowermost surface facing the base and coupled to an end of each electrode such that, as the plate changes in elevation with respect to the base, each electrode moves through its respective aperture, changing a position of each electrode with respect to the soil; placing a sample specimen inside a beaker; placing the soil inside the beaker and on the sample specimen; placing the testing apparatus on the soil with at least one electrode extending into the soil; and placing a weight on the plate to apply a load on the testing apparatus, allowing the testing apparatus to compact the soil under the load as each electrode detects a parameter of the sample specimen or the soil.

20. The method of claim 19, further comprising rotating the threaded studs as the testing apparatus compacts the soil, moving each electrode with respect to the base to maintain a desired distance between a tip of each electrode and the sample specimen.